Mississippi Valley State University

CS 341 — Discrete Structures

Lecture 18: Exactly One

Functions, Range, and Composition

Chartrand & Zhang, Section 5.3

Name: _____ Date: _____

Where we are

In a relation, an element of A may be related to all, some, or none of B. The case where each element of A is related to **exactly one** element of B is important enough to have a familiar name. Functions are a special kind of relation — so everything from last week still applies.

By the end of today you should be able to

1. Decide whether a given relation is a function.
2. Tell domain, codomain, and range apart.
3. Compute a composition and explain why order matters.

1. The definition

Definition 5.26

Let A and B be nonempty sets. A **function** f from A to B is a relation from A to B that associates with each element of A a **unique** element of B. We write $f : A \to B$.

So $f \subseteq A \times B$ with a restriction: every $a \in A$ is the first coordinate of **exactly one** pair.

Two ways a relation fails to be a function

A subset $g \subseteq A \times B$ is *not* a function from A to B if:

1. some $a \in A$ is the first coordinate of no pair — nothing to map it to;
2. some $a \in A$ is the first coordinate of more than one pair — the image is not unique.

Check both. Students who check only the second miss half the failures.

Test these. $A = \{1, 2, 3\}$, $B = \{a, b\}$.

Relation	Function?	Why
$\{(1,a),(2,b),(3,a)\}$	Yes	Each of $1,2,3$ appears exactly once.
$\{(1,a),(2,b)\}$	No	3 has no image.
$\{(1,a),(1,b),(2,a),(3,b)\}$	No	1 has two images.
$\{(1,b),(2,b),(3,b)\}$	Yes	Repeats in B are fine; only A is restricted.

2. Domain, codomain, range

Definition 5.28

For $f : A \to B$: A is the **domain**, B is the **codomain**, and the **range** is

$$f(A) = \{f(a) : a \in A\}.$$

Example 5.30. $A = \{1,2,3,4\}$, $B = \{a,b,c,d,e\}$, and $f = \{(1,b),(2,a),(3,b),(4,d)\}$.

Domain: $\{1,2,3,4\}$ Codomain: $\{a,b,c,d,e\}$ Range: $\{a,b,d\}$

Range and codomain are different things

The range is always a subset of the codomain, and the two are often *unequal* — here c and e are images of nothing.

The codomain is part of the function's definition, not something you compute. Two rules with the same formula but different codomains are different functions, and on Wednesday that difference will decide whether a function is onto.

Definition 5.29 — image of a subset

For $X \subseteq A$, $f(X) = \{f(x) : x \in X\}$.

With f as above: $f(\{1,2\}) = \{a,b\}$ and $f(\{1,3\}) = \{b\}$.

Note $|f(X)|$ can be *smaller* than $|X|$ — two inputs may share an image.

3. Composition

Composition

For $f : A \to B$ and $g : B \to C$, the **composition** $g \circ f : A \to C$ is defined by

$$(g \circ f)(x) = g\big(f(x)\big) \qquad \text{for all } x \in A.$$

Apply f first, then g — even though g is written on the left.

Example 5.42. $A = \{1, 2, 3\}$, $B = \{a, b, c, d\}$, $C = \{x, y, z\}$, with

$$f = \{(1, c), (2, b), (3, a)\}, \qquad g = \{(a, y), (b, x), (c, x), (d, z)\}.$$

$$(g \circ f)(1) = g(f(1)) = g(c) = x \quad (g \circ f)(2) = x \quad (g \circ f)(3) = y$$

So $g \circ f = \{(1, x), (2, x), (3, y)\}$.

4. Order matters

Let $f, g : \mathbb{R} \to \mathbb{R}$ with $f(x) = 2x + 1$ and $g(x) = x^2$.

$$(g \circ f)(x) = g(f(x)) = (2x + 1)^2 \qquad (f \circ g)(x) = f(g(x)) = 2x^2 + 1$$

At $x = 2$: $(g \circ f)(2) = 25$ while $(f \circ g)(2) = 9$.

Composition is not commutative

Even when both $f \circ g$ and $g \circ f$ are defined, they are usually different functions. You have met this shape three times now: $A - B \neq B - A$ for sets, reordering an `and` in code can crash it, and now composition. **Whenever an operation combines two things asymmetrically, check before assuming you may swap them.**

Composition in the wild

Let X be a set of students, $Y = \{A, B, C, D, F\}$ the possible grades, and $Z = \{0, 1, \ldots\}$ counts. If f assigns each student a grade and g assigns each grade the number of students receiving it, then $(g \circ f)(s)$ tells you how many students got the same grade as s.
That is Chartrand's Example 5.44, and it is a database join in miniature.

Exit ticket

1. Is $\{(1, a), (2, a)\}$ a function from $\{1, 2, 3\}$ to $\{a, b\}$? Explain.

2. $f, g : \mathbb{R} \to \mathbb{R}$ with $f(x) = x + 3$, $g(x) = x^2$. Find $(g \circ f)(1)$ and $(f \circ g)(1)$.

Before Wednesday

Read Section 5.4, "Bijective Functions," pp. 170–174 — one-to-one and onto.

Self-check (not collected). Section 5.3, Exercises 1, 3, and 13 — odd-numbered, answers in the back.

Homework 7 is due Friday, October 2, 11:59 p.m.

Sections 5.1 and 5.2. Problems 3 and 4 are proofs.

Questions I still have